MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website: mathsmanthan.co.in

Subject: Mathematics **WEEKLY CHAPTER TEST – 06 (SOLUTIONS)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 12: Surface Areas and Volumes** **MM: 20**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, factoring common terms like πr before substituting $\pi = \frac{22}{7}$ or 3.14, showing intermediate arithmetic clearly, and boxing the final answer with proper physical units ($\text{cm}^2, \text{cm}^3, \text{m}^2, \text{m}^3$).
2. Partial step marks are allocated in brackets (**[0.5 Mark]**, **[1 Mark]**, **[1.5 Marks]**, **[2 Marks]**). Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions & Assertion-Reason MATHS MANTHAN | (4 × 1 = 4 Marks)

Q1. The curved surface area of a cylinder is 264 m^2 and its volume is 924 m^3. The ratio of its diameter to its height is: [1]

(A) 3 : 7 (B) 7 : 3 (C) 6 : 7 (D) 7 : 6

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B) 7 : 3** [0.5 Mark]

Given: $\text{CSA} = 2\pi rh = 264\ \text{m}^2$, $V = \pi r^2 h = 924\ \text{m}^3$.

Formula / Theorem Used: $\dfrac{V}{\text{CSA}} = \dfrac{\pi r^2 h}{2\pi rh} = \dfrac{r}{2}$

$\frac{r}{2} = \frac{924}{264} = \frac{7}{2} \implies r = 7\ \text{m} \implies d = 14\ \text{m}$. From $2\left(\frac{22}{7}\right)(7)h = 264 \implies h = \frac{264}{44} = 6\ \text{m}$.

$$d : h = 14 : 6 = \boxed{7 : 3}$$

Q2. A cylinder with base radius of 8 cm and height of 2 cm is melted to form a cone of height 6 cm. The radius of the cone is: [1]

(A) 4 cm (B) 5 cm (C) 6 cm (D) 8 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(D) 8 cm** [0.5 Mark]

Given: Cylinder: $r_1 = 8$ cm, $h_1 = 2$ cm. Cone: $h_2 = 6$ cm, radius r_2.

Formula / Theorem Used: $\dfrac{1}{3}\pi r_2^2 h_2 = \pi r_1^2 h_1$

$$\frac{1}{3}r_2^2(6) = 8^2 \times 2 \implies 2r_2^2 = 2 \times 8^2 \implies r_2^2 = 64 \implies \boxed{\mathbf{r_2 = 8\ \text{cm}}}$$

Q3. If three metallic spheres of radii 6 cm, 8 cm, and 10 cm are melted to form a single solid sphere, the diameter of the resulting sphere is: [1]

(A) 12 cm (B) 24 cm (C) 30 cm (D) 36 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B) 24 cm** [0.5 Mark]

Given: $r_1 = 6$, $r_2 = 8$, $r_3 = 10$. Let new radius $= R$.

Formula / Theorem Used: $\dfrac{4}{3}\pi R^3 = \dfrac{4}{3}\pi(r_1^3 + r_2^3 + r_3^3)$

$R^3 = 6^3 + 8^3 + 10^3 = 216 + 512 + 1000 = 1728 = 12^3 \implies R = 12$ cm. $d = 2R = \boxed{24\ \text{cm}}$.

Q4. **Assertion (A):** If a solid cylinder, a solid cone, and a solid hemisphere stand on equal circular bases of radius r and have the same height r, the volume of the cylinder is equal to the sum of the volumes of the cone and the hemisphere.
Reason (R): If a cylinder, a cone, and a hemisphere have equal base radius r and height r, their volumes are in the ratio $3 : 1 : 2$. [1]

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A)** [0.5 Mark]

Verification: $V_{\text{cyl}} = \pi r^3$, $V_{\text{cone}} = \frac{1}{3}\pi r^3$, $V_{\text{hemi}} = \frac{2}{3}\pi r^3$.
Ratio $= 1 : \frac{1}{3} : \frac{2}{3} = 3 : 1 : 2$ (Reason (R) is true). Since $1 + 2 = 3$, $V_{\text{cone}} + V_{\text{hemi}} = V_{\text{cyl}}$, so both (A) and (R) are true and (R) explains (A).

SECTION B: Short Answer Questions

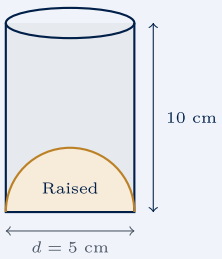
MATHS MANTHAN | **($2 \times 2 = 4$ Marks)**

Q5. A juice seller served customers using glasses of inner diameter 5 cm and height 10 cm. The bottom had a hemispherical raised portion reducing capacity. Find the apparent and actual capacity of the glass. (Use $\pi = 3.14$) [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Apparent Capacity ($\pi = 3.14$) [1 Mark]
Radius $r = \frac{5}{2} = 2.5$ cm, height $h = 10$ cm.

Formula / Theorem Used: Apparent $= \pi r^2 h$, Actual $= \pi r^2 h - \frac{2}{3}\pi r^3$



Apparent Capacity $= 3.14 \times (2.5)^2 \times 10 = 3.14 \times 62.5 =$ **196.25 cm^3**

Step 2: Actual Capacity [1 Mark]
Volume of raised hemisphere $= \frac{2}{3} \times 3.14 \times (2.5)^3 = \frac{2}{3} \times 3.14 \times 15.625 \approx 32.71$ cm^3.

Actual Capacity $= 196.25 - 32.71 =$ **163.54 cm^3**

Q6. A solid right circular cone of radius 1 cm stands on a hemisphere of radius 1 cm. If the height of the cone equals its radius, find the volume of the solid in terms of π. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Dimensions & Formula [1 Mark]
Given: Common radius $r = 1$ cm, height of cone $h = r = 1$ cm.

Formula / Theorem Used: $V_{\text{solid}} = V_{\text{cone}} + V_{\text{hemi}} = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$

Step 2: Substitution [1 Mark]

$$V_{\text{solid}} = \frac{1}{3}\pi(1)^2(1) + \frac{2}{3}\pi(1)^3 = \frac{\pi}{3} + \frac{2\pi}{3} = \boxed{\pi \text{ cm}^3}$$

Q7. A gulab jamun contains sugar syrup up to about 30% of its volume. Find approximately how much syrup would be found in 45 gulab jamuns, each shaped like a cylinder with two hemispherical ends of total length 5 cm and diameter 2.8 cm. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Dimensions of 1 Gulab Jamun [1 Mark]

Radius $r = \frac{2.8}{2} = 1.4$ cm; cylinder height $h = 5 - 2(1.4) = 2.2$ cm.

Formula / Theorem Used: $V_1 = \pi r^2 h + 2\left(\frac{2}{3}\pi r^3\right) = \pi r^2\left(h + \frac{4}{3}r\right)$

Step 2: Volume of 45 Gulab Jamuns [1 Mark]

$$V_1 = \frac{22}{7} \times 1.4 \times 1.4 \times \left(2.2 + \frac{5.6}{3}\right) = 6.16 \times \frac{12.2}{3} = \frac{75.152}{3} \text{ cm}^3$$

$V_{45} = 45 \times \frac{75.152}{3} = 15 \times 75.152 = 1127.28$ cm^3.

Step 3: 30% Sugar Syrup Volume [1 Mark]

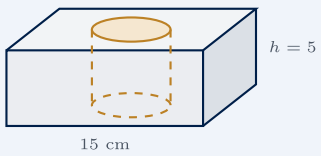
$$\text{Syrup} = \frac{30}{100} \times 1127.28 = 338.184 \text{ cm}^3 \approx \boxed{\mathbf{338 \text{ cm}^3}}$$

Q8. From a solid cuboidal metallic block of dimensions 15 cm × 10 cm × 5 cm, a cylindrical hole of diameter 7 cm is drilled out right through its thickness. Find the surface area of the remaining block. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Dimensions & Surface Accounting [1 Mark]

Cuboid: $l = 15$ cm, $b = 10$ cm, $h = 5$ cm (thickness). Hole radius $r = \frac{7}{2} = 3.5$ cm.



Formula / Theorem Used: $\text{TSA}_{\text{rem}} = 2(lb + bh + hl) - 2\pi r^2 + 2\pi rh$

Step 2: Cuboid TSA & Hole Adjustment [2 Marks]

$2(lb + bh + hl) = 2(150 + 50 + 75) = 2(275) = 550$ cm^2.

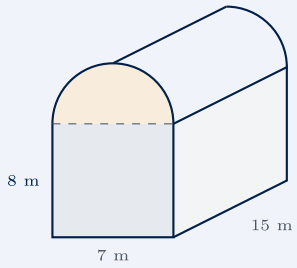
$2\pi r^2 = 2 \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} = 77$ cm^2; $2\pi rh = 2 \times \frac{22}{7} \times \frac{7}{2} \times 5 = 110$ cm^2.

$$\text{TSA}_{\text{rem}} = 550 - 77 + 110 = \boxed{\mathbf{583 \text{ cm}^2}}$$

Q9. Shanta runs an industry in a shed which is in the shape of a cuboid surmounted by a half-cylinder. The base of the shed is 7 m × 15 m, and the height of the cuboidal portion is 8 m.
(i) Find the volume of air that the empty shed can hold.
(ii) If the machinery inside occupies a total space of 300 m^3 and there are 20 workers each occupying 0.08 m^3, find the volume of air inside the shed. [6]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 6 Marks]

Given: Cuboid: $l = 15$ m, $b = 7$ m, $h = 8$ m. Half-cylinder: radius $r = \frac{7}{2} = 3.5$ m, length $L = 15$ m.



Step 1: (i) Volume of Empty Shed [3 Marks]

Formula / Theorem
Used: $V_{\text{shed}} = V_{\text{cuboid}} + \frac{1}{2}V_{\text{cylinder}} = lbh + \frac{1}{2}\pi r^2 L$

$V_{\text{cuboid}} = 15 \times 7 \times 8 = 840$ m^3.
$\frac{1}{2}V_{\text{cyl}} = \frac{1}{2} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 15 = \frac{11 \times 105}{4} = \frac{1155}{4} = 288.75$ m^3.

$$V_{\text{shed}} = 840 + 288.75 = \boxed{\mathbf{1128.75\ m^3}}$$

Step 2: (ii) Volume of Air Inside When Occupied [3 Marks]
Volume of machinery = 300 m^3; volume of 20 workers = $20 \times 0.08 = 1.6$ m^3.

$$V_{\text{air}} = 1128.75 - (300 + 1.6) = 1128.75 - 301.6 = \boxed{\mathbf{827.15\ m^3}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★